

Why SCDP is not used today?

- Because it is inefficient in terms of time, power consumption, utilization of modules and hardware cost
- Every instruction has same cycle length
- Clock period is determined by the slowest instruction (lw in our design – it uses 5 units) penalizing faster instruction
- Each functional unit can be used only once per clock; therefore, some functional units must be duplicated, raising the cost of the implementation
- All modules whether needed or not are working increasing the power consumption.

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- Circuit timings has to be considered properly.
- For complex systems e.g. processor involving floating point arithmetic- this doesn't work well at all.
- It violates the design principle “make common case fast” , as making the common case fast is useless, if this common case is not the worst case. Here worst case needs to be improved
- Modern processors use pipelining to overcome this.
- Overall performance is poor – but good for simple processors with small instruction sets.

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